Suggested problems: 1-10

Problem 1: Suppose X and Y are jointly continuous with probability density function

$$f(x,y) = \begin{cases} 6xy^2 & : 0 \le x \le 1,\ 0 \le y \le 1, \\ 0 & : \text{else.} \end{cases}$$

(a) Check whether X and Y are independent using marginal probability density functions.

(b) Check whether X and Y are independent using cumulative distribution functions.

Problem 2: Suppose X and Y are jointly continuous with probability density function

$$f(x,y) = \begin{cases} 2 & : 0 \le y \le x \le 1, \\ 0 & : \text{else.} \end{cases}$$

Are X and Y independent?

Problem 3: Suppose X and Y are jointly discrete random variables with joint probability function $f(x, y)$ given by

$f(x,y)$	$y=0$	$y=1$	$y=2$
$x=0$	0.38	0.14	0.24
$x=1$	0.17	0.02	0.05

Are X and Y independent?